Final Exam A: Cumulative

GSE 510

Total Points: 100
10 Questions, 10 Points Each

Instructions: Answer all questions. Show your work clearly. Each question is worth 10 points. Partial credit will be given for correct reasoning.

1. **Sets, Bounds, and Feasibility** **(10 points)**

Consider the set

$$B = \{(x_1, x_2) \in \mathbb{R}^2_+ : 3x_1 + 2x_2 \leq 18\}.$$

(a) Is $(2, 4) \in B$?

(b) Is $(5, 2) \in B$?

(c) Find the largest possible value of x_1.

(d) Find the largest possible value of x_2.

(e) Is B closed and bounded? Briefly explain.

2. **Logic and Sequences** **(10 points)**

Consider the statement:

$$\text{If } x > 3, \text{ then } x^2 > 9.$$

(a) Write the converse.

(b) Write the contrapositive.

(c) Is the original statement true?

(d) Is the converse true? Explain.

(e) Determine whether the sequence

$$a_n = 3 + \frac{5}{n}$$

converges. If it does, find its limit.

3. **Open and Closed Sets** **(10 points)**

Let

$$S = [1, 4).$$

(a) Is S open in $\mathbb{R}$?

(b) Is S closed in $\mathbb{R}$?

(c) Find the boundary points of S.

(d) Give a sequence in S that converges to a point not in S.

(e) Give a sequence in S that converges to a point in S.

4. **Linear Algebra and Systems** **(10 points)**

Consider the system

$$2x_1 + 3x_2 = 16, \qquad x_1 + 4x_2 = 14.$$

(a) Write the system in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Compute $|\mathbf{A}|$.

(c) Is $\mathbf{A}$ nonsingular?

(d) Solve for x_1 and x_2.

(e) Explain what nonsingularity means for this system.

5. **Span and Linear Independence** **(10 points)**

Let

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

(a) Write a general linear combination $a\mathbf{v}_1 + b\mathbf{v}_2$.

(b) Compute the determinant of the matrix whose columns are $\mathbf{v}_1$ and $\mathbf{v}_2$.

(c) Are the vectors linearly independent?

(d) Do the vectors span $\mathbb{R}^2$?

(e) Can they generate the vector

$$\begin{bmatrix} 7 \\ 8 \end{bmatrix}?$$

If yes, find a and b.

6. **OLS in Matrix Form** **(10 points)**

Suppose

$$\mathbf{y} = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}.$$

The regression model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}.$$

(a) Compute $\mathbf{X}'\mathbf{X}$.

(b) Compute $\mathbf{X}'\mathbf{y}$.

(c) Compute $(\mathbf{X}'\mathbf{X})^{-1}$.

(d) Compute $\widehat{\boldsymbol{\beta}}$.

(e) Write the fitted regression line.

7. **Differentiation, Approximation, and Interpretation** **(10 points)**

Suppose

$$C(q) = 100 + 12q + 2q^2.$$

(a) Find the marginal cost function.

(b) Compute $C'(10)$.

(c) Use a linear approximation to estimate $C(11) - C(10)$.

(d) Compute the exact value of $C(11) - C(10)$.

(e) Explain why the approximation is not exactly equal to the true change.

8. **Unconstrained Optimization with Two Variables** **(10 points)**

Consider

$$f(x,y) = 40x + 30y - 2x^2 - y^2 - xy.$$

(a) Compute f_x and f_y.

(b) Solve the first-order conditions.

(c) Compute the Hessian matrix.

(d) Use the Hessian to classify the critical point.

(e) Find the value of the function at the critical point.

9. **Differential Equation and Stability** **(10 points)**

Consider

$$\frac{dp}{dt} = 0.4(D(p) - S(p)),$$

where

$$D(p) = 90 - 2p, \qquad S(p) = 30 + p.$$

(a) Substitute demand and supply into the differential equation.

(b) Simplify the differential equation.

(c) Find the steady-state price.

(d) Determine whether the steady state is stable.

(e) Interpret the adjustment rule economically.

10. **Solow-Style Difference Equation** **(10 points)**

Consider the capital accumulation equation

$$k_{t+1} = sAk_t^{1/2} + (1-\delta)k_t,$$

where

$$s = 0.5, \qquad A = 8, \qquad \delta = 0.25.$$

(a) Substitute the parameter values into the equation.

(b) Find the steady state k^*.

(c) Suppose $k_0 = 64$. Compute k_1.

(d) Suppose $k_0 = 400$. Compute k_1.

(e) Based on parts (c) and (d), explain whether capital moves toward the steady state.

(f) Interpret the steady state economically.